Note on Axion Metal

Kai-Zhi Bai

Department of Physics, The University of Hong Kong,

Pokfulam Road, Hong Kong, China

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Abstract

We derive the axion electrodynamics under SI unit. We show side-surface half quantized anomalous Hall effect in axion insulator, and term the effect axion metal.

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I. Axion Electrodynamics

Following [1, 2], we re-derive the axion electrodynamics in SI unit. The equations read

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0} - \frac{\alpha}{\pi} c \nabla\theta \cdot \boldsymbol{B}, \tag{1a}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{1b}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{1c}$$

$$\nabla \times \boldsymbol{B} = \frac{1}{c^2}\partial_t \boldsymbol{E} + \mu_0 \boldsymbol{J} + \frac{\alpha}{\pi c}(\partial_t \theta \boldsymbol{B} + \nabla\theta \times \boldsymbol{E}), \tag{1d}$$

with $\alpha = \dfrac{e^2}{4\pi\varepsilon_0 \hbar c}$ as the fine structure constant. In the section, we shall derive the dynamics using Euler-Lagrange equation, with correct coefficients under SI units.

We start from extracting the correct form of the Theta term[2]

$$S_\theta = \int \mathrm{d}^4 x \frac{\theta e^2}{4\pi^2 \hbar} \boldsymbol{E} \cdot \boldsymbol{B}. \tag{2}$$

Its dimension is $[S_\theta] = \dfrac{s \cdot m^3}{J \cdot s}\left[\dfrac{(eE)^2}{c}\right] = \dfrac{m^3}{J}\dfrac{[F]^2}{m/s} = \dfrac{m^2 \cdot s}{J}\dfrac{J^2}{m^2} = J \cdot s$, which is correct as an action. We have utilized Faraday's law to transform dimension of B into E/c. To derive the dynamics, we need to rewrite the action using 4-vector notation with electromagnetic vector field. Following the $(+1, -1, -1, -1)$ metric convention, we let

$$x^\mu = (x^0, \boldsymbol{x}),\ x^0 = ct, \tag{3a}$$

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\partial_t, \nabla\right), \tag{3b}$$

$$A^\mu = (A^0, \boldsymbol{A}),\ A^0 = \phi/c, \tag{3c}$$

$$J^\mu = (c\rho, \boldsymbol{J}). \tag{3d}$$

We shall also distinguish the upper and lower indices of electric and magnetic field, such that

$$\begin{cases} \boldsymbol{E} = -\nabla\varphi - \partial_t \boldsymbol{A} \\ \boldsymbol{B} = \nabla \times \boldsymbol{A} \end{cases} \Longrightarrow \begin{cases} E^i = c(\partial^i A^0 - \partial^0 A^i) \\ B_i = -\varepsilon_{ijk}\partial^j A^k \end{cases}. \tag{4}$$

Notice that we have introduced the Levi-Civita symbol in Minkowski space with $\varepsilon^{123} = 1$ and $\varepsilon_{ijk} = -\varepsilon^{ijk}$ by lowering the index. It is indeed part of $\varepsilon^{\mu\nu\rho\lambda}$ with $\varepsilon^{0123} = 1$. We then define the antisymmetric electromagnetic field tensor

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu, \tag{5}$$

and find that

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = -\frac{1}{c}E_i, \tag{6a}$$

$$F^{i0} = \partial^i A^0 - \partial^0 A^i = \frac{1}{c}E^i, \tag{6b}$$

$$\frac{1}{2}\varepsilon^{ijk}F_{jk} = \varepsilon^{ijk}\partial_j A_k = -B^i. \tag{6c}$$

Now we consider a combination

$$\begin{aligned}
\varepsilon^{\mu\nu\rho\lambda}F_{\mu\nu}F_{\rho\lambda} &= \varepsilon^{\mu\nu\rho\lambda}(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial_\rho A_\lambda - \partial_\lambda A_\rho) \\
&= 4\varepsilon^{\mu\nu\rho\lambda}\partial_\mu A_\nu \partial_\rho A_\lambda \\
&= 4\left(\varepsilon^{0\nu\rho\lambda}\partial_0 A_\nu \partial_\rho A_\lambda + \varepsilon^{\mu 0\rho\lambda}\partial_\mu A_0 \partial_\rho A_\lambda + (\rho = 0) + (\lambda = 0)\right) \\
&= 8\varepsilon^{0ijk}(\partial_0 A_i - \partial_i A_0)\partial_j A_k \\
&= 8E_i B^i/c = 8\boldsymbol{E}\cdot\boldsymbol{B}/c.
\end{aligned} \tag{7}$$

This enlightens us to rewrite the θ-action as

$$S_\theta = \int \mathrm{d}^4x \frac{\theta e^2 c}{32\pi^2\hbar}\varepsilon^{\mu\nu\rho\lambda}F_{\mu\nu}F_{\rho\lambda}, \tag{8}$$

which is a 4-form integral and is special in (3+1)-spacetime.

The Lagrangian density of Maxwell-axion electrodynamics

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_{\text{Maxwell}} + \mathcal{L}_{\text{axion}} \\
&= -\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu + \kappa\theta\varepsilon^{\mu\nu\rho\lambda}F_{\mu\nu}F_{\rho\lambda},
\end{aligned} \tag{9}$$

where we have introduced the axion coupling constant $\kappa = \dfrac{e^2 c}{32\pi^2\hbar}$. We may check that the dimension of the Lagrangian is correct since for the field term, $s\cdot m^3\left[\dfrac{F_{\mu\nu}^2}{\mu_0}\right] = s\cdot m^3\left[\dfrac{B^2}{\mu_0}\right] = s\cdot m^3\left[\dfrac{E^2}{c^2\mu_0}\right] = s\cdot m^3\left[E^2\varepsilon_0\right] = s\cdot m^3\left[\dfrac{(eE)^2}{Fr^2}\right] = s\cdot m[F] = J\cdot s$, while for the field-matter coupling term, $s\cdot m^3[\rho\varphi] = s[e\varphi] = J\cdot s$. Now, the Euler-Lagrange equation reads

$$\begin{aligned}
0 = \frac{\delta\mathcal{L}}{\delta A_\alpha} &= \frac{\partial\mathcal{L}}{\partial A_\alpha} - \partial_\beta\frac{\partial\mathcal{L}}{\partial(\partial_\beta A_\alpha)} \\
&= -J^\mu\frac{\partial A_\mu}{\partial A_\alpha} - \partial_\beta\frac{\partial}{\partial(\partial_\beta A_\alpha)}\left[-\frac{1}{2\mu_0}\partial^\mu A^\nu(\partial_\mu A_\nu - \partial_\nu A_\mu) - 4\kappa\theta\varepsilon^{\mu\nu\rho\lambda}\partial_\mu A_\nu\partial_\rho A_\lambda\right] \\
&= -J^\alpha + \frac{1}{\mu_0}\partial_\beta(\partial^\beta A^\alpha - \partial^\alpha A^\beta) - 8\kappa\varepsilon^{\alpha\beta\rho\lambda}\partial_\beta(\theta\partial_\rho A_\lambda).
\end{aligned} \tag{10}$$

Notice that $8\kappa\mu_0 = \dfrac{\theta e^2}{4\pi^2\varepsilon_0\hbar} = \dfrac{\alpha}{\pi}$ is the fine structure constant, and we thus write

$$\partial_\beta F^{\beta\alpha} = \partial_\beta(\partial^\beta A^\alpha - \partial^\alpha A^\beta) = \mu_0 J^\alpha + \frac{\alpha}{\pi}\varepsilon^{\alpha\beta\rho\lambda}\partial_\beta(\theta\partial_\rho A_\lambda). \tag{11}$$

This is the axion electrodynamics, which is modified by the last term compared with the Maxwell equations. For components, we first set $\alpha = 0$, and find

$$\begin{aligned} \partial_i(\partial^i A^0 - \partial^0 A^i) &= \mu_0 J^0 + \frac{\alpha}{\pi}\varepsilon^{0ijk}\partial_i(\theta\partial_j A_k), \\ \Longrightarrow \frac{1}{c}\partial_i E^i &= \mu_0 c\rho - \frac{\alpha}{\pi}(\partial_i\theta)B^i, \\ \Longrightarrow \nabla\cdot\boldsymbol{E} &= \frac{\rho}{\varepsilon_0} - \frac{\alpha}{\pi}c\nabla\theta\cdot\boldsymbol{B}, \end{aligned} \tag{12}$$

as the modified Gauss's law for electric field. We then set $\alpha = i$, and find

$$\partial_\beta(\partial^\beta A^i - \partial^i A^\beta) = \mu_0 J^i + \frac{\alpha}{\pi}\varepsilon^{i\beta\rho\lambda}\partial_\beta(\theta\partial_\rho A_\lambda). \tag{13}$$

We notice that the left-hand side can be split into

$$\begin{aligned} \partial_\beta(\partial^\beta A^i - \partial^i A^\beta) &= \partial_0(\partial^0 A^i - \partial^i A^0) + \partial_j(\partial^j A^i - \partial^i A^j) \\ &= -\frac{1}{c}\partial_0 E^i + (\nabla\times\boldsymbol{B})^i = -\frac{1}{c^2}\partial_t\boldsymbol{E}^i + (\nabla\times\boldsymbol{B})^i, \end{aligned} \tag{14}$$

where the last equality is obtained by observing that

$$\begin{aligned} (\nabla\times\boldsymbol{B})^i &= -\varepsilon^{ijk}\partial_j B_k = \varepsilon^{ijk}\partial_j\varepsilon_{klm}\partial^l A^m \\ &= -(\delta^i_l\delta^j_m - \delta^i_m\delta^j_l)\partial_j\partial^l A^m = -\partial_j\partial^i A^j + \partial_j\partial^j A^i. \end{aligned} \tag{15}$$

For the second term on the right-hand side, we have

$$\begin{aligned} \varepsilon^{i\beta\rho\lambda}\partial_\beta(\theta\partial_\rho A_\lambda) &= \varepsilon^{i\beta\rho\lambda}(\partial_\beta\theta)\partial_\rho A_\lambda \\ &= \varepsilon^{i0jk}(\partial_0\theta)\partial_j A_k + \varepsilon^{ij0k}(\partial_j\theta)(\partial_0 A_k - \partial_k A_0) \\ &= (\partial_0\theta)B^i - \frac{1}{c}\varepsilon^{ijk}\partial_j\theta E_k \\ &= \frac{1}{c}(\partial_t\theta)\boldsymbol{B}^i + \frac{1}{c}(\nabla\theta\times\boldsymbol{E})^i. \end{aligned} \tag{16}$$

Finally, we arrive at the modified Ampere's law

$$\nabla\times\boldsymbol{B} = \frac{1}{c^2}\partial_t\boldsymbol{E} + \mu_0\boldsymbol{J} + \frac{\alpha}{\pi c}(\boldsymbol{B}\partial_t\theta + \nabla\theta\times\boldsymbol{E}). \tag{17}$$

Notice that the Bianchi identity $\partial_\alpha\tilde{F}^{\alpha\beta} = 0$, where the Hodge dual tensor $\tilde{F}^{\alpha\beta} = \dfrac{1}{2}\varepsilon^{\alpha\beta\mu\nu}F_{\mu\nu}$, is unmodified by the axion term. This identity offers the non-divergence Gauss's law for magnetic field and Faraday's law, as for $\beta = 0$,

$$0 = \partial_i\varepsilon^{i0jk}F_{jk} \Longrightarrow \nabla\cdot\boldsymbol{B} = 0, \tag{18}$$

while for $\beta = i$,

$$\begin{aligned} 0 &= \partial_\alpha \varepsilon^{\alpha i \mu\nu} F_{\mu\nu} = \partial_0 \varepsilon^{0ijk} F_{jk} + 2\partial_j \varepsilon^{ji0k} F_{0k} \\ &= -2\partial_0 B^i - \frac{2}{c}\varepsilon^{ijk}\partial_j E_k, \end{aligned} \tag{19}$$

i.e.,

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}. \tag{20}$$

Equations (12), (17), (18), and (20) are the axion electrodynamics in SI unit.

We define a new electromagnetic tensor

$$\mathcal{F}^{\mu\nu} = F^{\mu\nu} + \alpha\frac{\theta}{\pi}\varepsilon^{\mu\nu\rho\lambda}\partial_\rho A_\lambda, \tag{21}$$

which satisfies from Eq. (11)

$$\partial_\alpha \mathcal{F}^{\alpha\beta} = \mu_0 J^\beta. \tag{22}$$

The new Hodge dual tensor reads

$$\begin{aligned} \tilde{\mathcal{F}}^{\mu\nu} &= \frac{1}{2}\varepsilon^{\mu\nu\rho\lambda}\mathcal{F}_{\rho\lambda} = \tilde{F}^{\mu\nu} + \alpha\frac{\theta}{2\pi}\varepsilon^{\mu\nu\rho\lambda}\varepsilon_{\rho\lambda\alpha\beta}\partial^\alpha A^\beta \\ &= \tilde{F}^{\mu\nu} + \alpha\frac{\theta}{\pi}\delta^{\mu\nu}_{\alpha\beta}\partial^\alpha A^\beta = \tilde{F}^{\mu\nu} + \alpha\frac{\theta}{\pi}F^{\mu\nu}, \end{aligned} \tag{23}$$

and the Bianchi identity is modified as

$$\partial_\alpha \tilde{\mathcal{F}}^{\alpha\beta} = \alpha\frac{\theta}{\pi}\partial_\alpha F^{\alpha\beta}. \tag{24}$$

By the modified Gauss's law Eq. (1a) and Ampere's law Eq. (1d), we may redefine the new electric and magnetic fields as

$$\tilde{\boldsymbol{E}} = \boldsymbol{E} + \alpha c\frac{\theta}{\pi}\boldsymbol{B}, \tag{25a}$$

$$\tilde{\boldsymbol{B}} = \boldsymbol{B} - \frac{\alpha}{c}\frac{\theta}{\pi}\boldsymbol{E}, \tag{25b}$$

which can be written more compactly as

$$\begin{pmatrix} \tilde{\boldsymbol{E}}/c \\ \tilde{\boldsymbol{B}} \end{pmatrix} = \begin{pmatrix} 1 & \alpha\frac{\theta}{\pi} \\ -\alpha\frac{\theta}{\pi} & 1 \end{pmatrix} \begin{pmatrix} \boldsymbol{E}/c \\ \boldsymbol{B} \end{pmatrix} = \left(1 + i\alpha\frac{\theta}{\pi}\sigma_y\right) \begin{pmatrix} \boldsymbol{E}/c \\ \boldsymbol{B} \end{pmatrix}, \tag{26}$$

and Inversely, we have

$$\begin{pmatrix} \boldsymbol{E}/c \\ \boldsymbol{B} \end{pmatrix} = \frac{1}{1 + \alpha^2\theta^2/\pi^2}\left(1 - i\alpha\frac{\theta}{\pi}\sigma_y\right) \begin{pmatrix} \tilde{\boldsymbol{E}}/c \\ \tilde{\boldsymbol{B}} \end{pmatrix}. \tag{27}$$

The equations of motion in terms of the new fields read, for the Gauss's law of the electric field,

$$\nabla \cdot \tilde{\boldsymbol{E}} = \frac{\rho}{\varepsilon_0} - \alpha c \frac{\theta}{\pi} \nabla \cdot \boldsymbol{B} = \frac{\rho}{\varepsilon_0}; \tag{28}$$

for the Ampere's law of the magnetic field,

$$\begin{aligned} \nabla \times \tilde{\boldsymbol{B}} &= \frac{1}{c^2}\partial_t \boldsymbol{E} + \mu_0 \boldsymbol{J} + \frac{\alpha}{c}\left[\partial_t\left(\frac{\theta}{\pi}\right)\boldsymbol{B} - \frac{\theta}{\pi}\nabla \times \boldsymbol{E}\right] \\ &= \frac{1}{c^2}\partial_t \boldsymbol{E} + \mu_0 \boldsymbol{J} + \frac{\alpha}{c}\left[\partial_t\left(\frac{\theta}{\pi}\right)\boldsymbol{B} + \frac{\theta}{\pi}\partial_t \boldsymbol{B}\right] \\ &= \frac{1}{c^2}\partial_t\left(\boldsymbol{E} + \alpha c \frac{\theta}{\pi}\boldsymbol{B}\right) + \mu_0 \boldsymbol{J} \\ &= \frac{1}{c^2}\partial_t \tilde{\boldsymbol{E}} + \mu_0 \boldsymbol{J}; \end{aligned} \tag{29}$$

for the Gauss's law of the magnetic field,

$$\begin{aligned} \nabla \cdot \tilde{\boldsymbol{B}} &= \nabla \cdot \boldsymbol{B} - \frac{\alpha}{c}\left[\nabla\left(\frac{\theta}{\pi}\right)\cdot \boldsymbol{E} + \frac{\theta}{\pi}\nabla \cdot \boldsymbol{E}\right] \\ &= -\frac{\alpha}{c}\left[\nabla\left(\frac{\theta}{\pi}\right)\cdot \boldsymbol{E} + \frac{\theta}{\pi}\left(\frac{\rho}{\varepsilon_0} - \frac{\alpha}{\pi}c\nabla\theta \cdot \boldsymbol{B}\right)\right] \\ &= -\frac{\alpha}{c}\left[\nabla\left(\frac{\theta}{\pi}\right)\cdot\left(\boldsymbol{E} - \alpha c\frac{\theta}{\pi}\boldsymbol{B}\right) + \frac{\theta}{\pi}\frac{\rho}{\varepsilon_0}\right] \\ &= -\frac{\alpha}{c}\left[\frac{1}{1+\alpha^2\theta^2/\pi^2}\nabla\left(\frac{\theta}{\pi}\right)\cdot\left((1+\alpha^2\theta^2/\pi^2)\tilde{\boldsymbol{E}} - 2\alpha c\frac{\theta}{\pi}\tilde{\boldsymbol{B}}\right) + \frac{\theta}{\pi}\frac{\rho}{\varepsilon_0}\right] \\ &= -\frac{\alpha}{c}\left[\nabla\left(\frac{\theta}{\pi}\right)\cdot\left(\tilde{\boldsymbol{E}} - \frac{2\alpha c\theta/\pi}{1+\alpha^2\theta^2/\pi^2}\tilde{\boldsymbol{B}}\right) + \frac{\theta}{\pi}\frac{\rho}{\varepsilon_0}\right] \end{aligned} \tag{30}$$

for the Faraday's law of the electric field,

$$\begin{aligned} \nabla \times \tilde{\boldsymbol{E}} &= \nabla \times \boldsymbol{E} + \alpha c\left[\nabla\left(\frac{\theta}{\pi}\right)\times \boldsymbol{B} + \frac{\theta}{\pi}\nabla \times \boldsymbol{B}\right] \\ &= -\frac{\partial \boldsymbol{B}}{\partial t} + \alpha c\left[\nabla\left(\frac{\theta}{\pi}\right)\times \boldsymbol{B} + \frac{\theta}{\pi}\left(\frac{1}{c^2}\partial_t \boldsymbol{E} + \mu_0 \boldsymbol{J} + \frac{\alpha}{\pi c}(\partial_t \theta \boldsymbol{B} + \nabla\theta \times \boldsymbol{E})\right)\right] \\ &= -\frac{\partial \tilde{\boldsymbol{B}}}{\partial t} + \alpha c\left[\nabla\left(\frac{\theta}{\pi}\right)\times \boldsymbol{B} - \frac{1}{c^2}\partial_t\left(\frac{\theta}{\pi}\right)\boldsymbol{E} + \frac{\theta}{\pi}\left(\mu_0 \boldsymbol{J} + \frac{\alpha}{\pi c}(\partial_t\theta \boldsymbol{B} + \nabla\theta \times \boldsymbol{E})\right)\right] \\ &= -\frac{\partial \tilde{\boldsymbol{B}}}{\partial t} + \alpha c\left[\nabla\left(\frac{\theta}{\pi}\right)\times\left(\boldsymbol{B} + \frac{\alpha}{c}\frac{\theta}{\pi}\boldsymbol{E}\right) - \frac{1}{c^2}\partial_t\left(\frac{\theta}{\pi}\right)\left(\boldsymbol{E} - \alpha c\frac{\theta}{\pi}\boldsymbol{B}\right) + \frac{\theta}{\pi}\mu_0 \boldsymbol{J}\right] \\ &= \alpha c\left[\nabla\left(\frac{\theta}{\pi}\right)\times\left(\tilde{\boldsymbol{B}} - \frac{2\alpha\theta/c\pi}{1+\alpha^2\theta^2/\pi^2}\tilde{\boldsymbol{E}}\right) - \frac{1}{c^2}\partial_t\left(\frac{\theta}{\pi}\right)\left(\tilde{\boldsymbol{E}} - \frac{2\alpha c\theta/\pi}{1+\alpha^2\theta^2/\pi^2}\tilde{\boldsymbol{B}}\right)\right] \\ &\quad - \frac{\partial \tilde{\boldsymbol{B}}}{\partial t} + \alpha c\frac{\theta}{\pi}\mu_0 \boldsymbol{J}. \end{aligned} \tag{31}$$

II. From the Theta term to the Chern-Simons term

For a correctly regulated TI surface, the axion term can be rewritten as a Chern-Simons term at the surface. Consider the x-direction is opened, and the surface is at $x = 0$,

$$\begin{aligned} S_\theta &= \int \mathrm{d}^4x \frac{\theta e^2 c}{32\pi^2\hbar}\varepsilon^{\mu\nu\rho\lambda}F_{\mu\nu}F_{\rho\lambda} = \int \mathrm{d}^4x \frac{\theta e^2 c}{8\pi^2\hbar}\varepsilon^{\mu\nu\rho\lambda}\partial_\mu A_\nu \partial_\rho A_\lambda \\ &= \frac{e^2c^2}{4\pi h}\int \mathrm{d}t\mathrm{d}y\mathrm{d}z \int \mathrm{d}x \varepsilon^{\mu\nu\rho\lambda}(\partial_\mu(\theta A_\nu \partial_\rho A_\lambda) - (\partial_\mu\theta)A_\nu\partial_\rho A_\lambda - \theta A_\nu \partial_\mu\partial_\rho A_\lambda) \\ &= -\frac{e^2c^2}{4\pi h}\int \mathrm{d}t\mathrm{d}y\mathrm{d}z \int \mathrm{d}x \varepsilon^{\mu\nu\rho\lambda}\delta^x_\mu[-\delta(x)](2n+1)\pi A_\nu\partial_\rho A_\lambda \\ &= -\frac{e^2}{2h}(n+\frac{1}{2})\int \mathrm{d}^3x \varepsilon^{\nu\rho\lambda}cA_\nu\partial_\rho A_\lambda, \end{aligned} \tag{32}$$

which is the (2 + 1)D Chern-Simons term at the surface, and predicts the half-quantized anomalous Hall effect at the surface.

III. Dual Fields Generation, and Charge and Current Accumulation at the Surface

A. Gapless surface with half quantized Hall conductance

We now propose a general mechanism for the dual fields generation, and charge and current accumulation at the surfaces of a topological insulator (TI), as illustrated in Fig. 1. We would like to combine the axion electrodynamics with the half-quantized anomalous Hall effect in a semi-magnetic TI, and illustrate how the Zeeman field direction on the gapped surface determines the sign of the θ field in the TI thus the sign of Hall conductance. We also show that the charge and current accumulation can be separately described by the (2+1)D Chern-Simons theory at the gapped and the gapless surface, respectively.

We consider the stationary axion electrodynamics as

$$\nabla \cdot \tilde{\boldsymbol{E}} = \frac{\rho}{\varepsilon_0}, \tag{33a}$$

$$\nabla \times \boldsymbol{E} = 0, \tag{33b}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{33c}$$

$$\nabla \times \tilde{\boldsymbol{B}} = \mu_0 \boldsymbol{J}, \tag{33d}$$

$$\tilde{\boldsymbol{E}} = \boldsymbol{E} + \alpha\nu\boldsymbol{B}, \tag{33e}$$

$$\tilde{\boldsymbol{B}} = \boldsymbol{B} - \alpha\nu\boldsymbol{E}, \tag{33f}$$

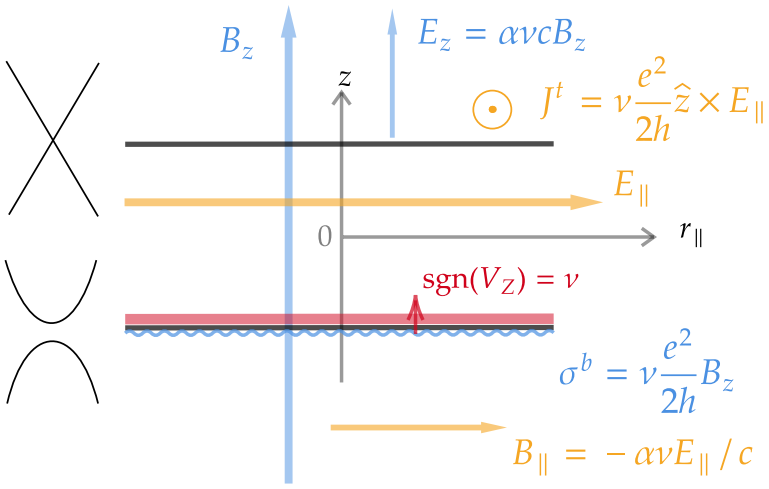


FIG. 1: Schematic diagram of the dual fields generation, and charge and current accumulation at the surfaces of a topological insulator. The different behaviors depend on whether the surface is gapped or gapless. The θ field in TI is defined by $\nu = \theta/\pi$, and is determined by the Zeeman field direction. Basically, an applied out-of-film magnetic field generates an electric field outside the gapless surface, and electric charges at the gapped surface. Oppositely, an in-film electric field generates a magnetic field outside the gapped surface, and magnetization current at the gapless surface.

where we have defined $\nu = \theta/\pi$ and absorbed a $1/c$ factor into $\boldsymbol{E}$. In our setup, we always choose the vacuum to own $\nu = 0$ as a trivial one. We choose the semi-magnetic TI film to lie at $[-L/2, L/2]$ along z, with its bottom surface gapped out by a Zeeman field polarized along z-direction, and the top surface remains gapless. We take x-y plane to be periodic such that the boundary does not affect. Due to the gapped nature of the bottom surface, there can not be current flowing in the surface but only charges accumulated, i.e., $\boldsymbol{j}^b = 0$, $\sigma^b \neq 0$; oppositely, due to the gapless nature of the top surface, there can not be charges accumulated but only current flowing, i.e., $\boldsymbol{j}^t \neq 0$, $\sigma^t = 0$.

Consider the Faraday's law Eq. (33b) and magnetic Gauss's law Eq. (33c), and we may take long-thin loop or large-thin cylinder for each case to illustrate that B_z and $\boldsymbol{E}_\parallel$ have to be constant throughout the whole space, through their integral formula with the Stokes' theorem. We can thus take these two fields as the probing source fields.

Firstly, we consider adding the source magnetic field $\boldsymbol{B} = B_z\hat{z}$ along the z-direction, which will involve the modified Gauss's law Eq. (33a). To see what happens at the boundaries,

we consider a large-thin cylinder with its axis along the z-direction, whose radius is R and thickness 2ε sitting at $z = \pm L/2$. An integrated modified Gauss's law in this volume leads to

$$\begin{aligned} &\iiint \mathrm{d}V \nabla \cdot \tilde{\boldsymbol{E}} = \iiint \mathrm{d}V \frac{\rho}{\varepsilon_0} \\ \Longrightarrow &\oiint \mathrm{d}\boldsymbol{S} \cdot \tilde{\boldsymbol{E}} = \frac{\pi R^2}{\varepsilon_0} \int_{z-\varepsilon}^{z+\varepsilon} \mathrm{d}z\rho \\ \Longrightarrow &\Delta \tilde{E}_z(z) = \frac{\sigma(z)}{\varepsilon_0}, \end{aligned} \tag{34}$$

where we have taken $\varepsilon \to 0$ and defined a surface charge density $\sigma(z) = \int_{z-\varepsilon}^{z+\varepsilon} \mathrm{d}z\rho$. Let us consider the two surfaces, and by $\tilde{E}_z = E_z + \alpha\nu B_z$ we have

$$\Delta E_z^t = \alpha\nu B_z, \tag{35a}$$

$$\Delta E_z^b = -\alpha\nu B_z + \frac{\sigma^b}{\varepsilon_0}, \tag{35b}$$

where we have utilized the fact that $\sigma^t = 0$ and $\sigma^b \neq 0$. Assuming that the electric field in the TI bulk vanishes (equivalent to vanishing bulk charge density by $\sigma^t = 0$), we take $E_z(|z| < L/2) = 0$, and thus obtain

$$E_z(z > L/2) = \alpha\nu B_z, \tag{36}$$

as the generated electric field outside the gapless surface. For the bottom surface, we notice that the gapped bottom surface serves as a natural electric capacitance, and from the perspective of minimal energy, the generation of an electric field should be generally avoided, thus we have $E_z(z < -L/2) = 0$ but an accumulated charge density

$$\sigma^b = -\alpha\nu c B_z \varepsilon_0 = -\nu \frac{e^2}{2h} B_z, \tag{37}$$

with physical constants restored.

Next we consider adding the source electric field $\boldsymbol{E} = \boldsymbol{E}_\parallel$ with only in-film plane component, which will involve the modified Ampere's law Eq. (33d). To consider its boundary physics, we take a long-thin loop with its long direction parallel with $\boldsymbol{E}_\parallel$, whose length is L and thickness 2ε sitting at $z = \pm L/2$. We take a positive boundary orientation such that the top edge has an opposite orientation with $\boldsymbol{E}_\parallel$. An integrated modified

Ampere's law in this area leads to

$$\begin{aligned} &\iint \mathrm{d}\boldsymbol{S}\cdot\nabla\times\tilde{\boldsymbol{B}}=\mu_0\iint \mathrm{d}\boldsymbol{S}\cdot\boldsymbol{J}\\ \Longrightarrow&\oint \mathrm{d}\boldsymbol{l}\cdot\tilde{\boldsymbol{B}}=\mu_0 L\int_{z-\varepsilon}^{z+\varepsilon}\mathrm{d}z\boldsymbol{J}\cdot(\hat{\boldsymbol{E}}_{\parallel}\times\hat{z})\\ \Longrightarrow&\hat{z}\times\Delta\tilde{\boldsymbol{B}}_{\parallel}(z)=\mu_0\boldsymbol{j}(z), \end{aligned} \tag{38}$$

where we have taken $\varepsilon\to 0$ and defined a surface current density $\boldsymbol{j}(z)=\int_{z-\varepsilon}^{z+\varepsilon}\mathrm{d}z\boldsymbol{J}$. The direction can be easily verified by taking some particular cases. Let us consider the two surfaces, and by $\tilde{\boldsymbol{B}}_{\parallel}=\boldsymbol{B}_{\parallel}-\alpha\nu\boldsymbol{E}_{\parallel}$ we have

$$\hat{z}\times\Delta\boldsymbol{B}_{\parallel}^{t}+\hat{z}\times\alpha\nu\boldsymbol{E}_{\parallel}=\mu_0\boldsymbol{j}^{t}, \tag{39a}$$

$$\hat{z}\times\Delta\boldsymbol{B}_{\parallel}^{b}-\hat{z}\times\alpha\nu\boldsymbol{E}_{\parallel}=0, \tag{39b}$$

where we have utilized the fact that $\boldsymbol{j}^{b}=0$ and $\boldsymbol{j}^{t}\neq 0$. Assuming that $\boldsymbol{B}_{\parallel}=0$ in TI bulk (equivalent to vanishing current density in the bulk by $\boldsymbol{j}^{b}=0$), we obtain

$$\boldsymbol{B}_{\parallel}(z<-L/2)=-\alpha\nu\boldsymbol{E}_{\parallel}/c, \tag{40}$$

as a generated parallel magnetic field outside the gapped surface. Similarly, by energy minimization, we have $\boldsymbol{B}_{\parallel}(z>L/2)=0$ but a magnetization current density

$$\boldsymbol{j}^{t}=\alpha\nu\hat{z}\times\boldsymbol{E}_{\parallel}/\mu_0=\nu\frac{e^2}{2h}\hat{z}\times\boldsymbol{E}_{\parallel}, \tag{41}$$

where in the last equality we restored the physical constants and utilize that $\dfrac{\alpha}{c\mu_0}=\alpha c\varepsilon_0=\dfrac{e^2}{4\pi\varepsilon_0\hbar c}c\varepsilon_0=\dfrac{e^2}{4\pi\hbar}=\dfrac{e^2}{2h}$. This formula gives us the desired half-quantized anomalous Hall effect at the gapless surface, and the sign of the Hall conductance is determined by the sign of ν. Furthermore, we know that this Hall conductance solely depends on the sign of the Zeeman field, thus we relate

$$\nu=\mathrm{sgn}(V_Z), \tag{42}$$

and write

$$\sigma_H=\frac{e^2}{2h}\mathrm{sgn}(V_Z). \tag{43}$$

The Chern-Simons theory reads as

$$S_{CS}=-(-1)^{p}\nu\frac{e^2}{4h}\int \mathrm{d}^3x\varepsilon^{\nu\rho\lambda}cA_\nu\partial_\rho A_\lambda, \tag{44}$$

where $p = t/b = 0/1$, and the current formula is readily obtained as

$$j^{\alpha} = \frac{\delta S_{CS}}{\delta A_{\alpha}} = -(-1)^{p}\nu\frac{e^{2}}{2h}c\varepsilon^{\alpha\rho\lambda}\partial_{\rho}A_{\lambda}, \tag{45}$$

which gives the time and spatial components of the surface current as

$$\sigma^{b} = j^{0,b}/c = \nu\frac{e^{2}}{2h}\varepsilon^{0ij}\partial_{i}A_{j} = -\nu\frac{e^{2}}{2h}B_{z}, \tag{46a}$$

$$\boldsymbol{j}^{t} = \nu\frac{e^{2}}{2h}\hat{z}\times\boldsymbol{E}_{\parallel}. \tag{46b}$$

Notice that in this way, the Chern-Simons theory is an effective theory, which stands depending on possible physical realization, i.e., depending on whether the surface is gapped or gapless. When the surface is gapped, the theory predicts a charge accumulation at the surface, while when the surface is gapless, the theory predicts a current flow at the surface.

Another thing to notice is since on the gapless surface, the parallel electric field will also generate a longitudinal current, the surface current indeed shares a non-$\pi/2$ angle with the electric field

$$\boldsymbol{j}^{t} = \boldsymbol{j}^{t}_{longi} + \boldsymbol{j}^{t}_{trans} = \sigma_{\parallel}\boldsymbol{E}_{\parallel} + \nu\frac{e^{2}}{2h}\hat{z}\times\boldsymbol{E}_{\parallel}, \tag{47}$$

with a Hall angle

$$\tan\varphi = \nu\frac{e^{2}}{2h\sigma_{\parallel}}, \tag{48}$$

and by considering a minimal conductance of the Dirac cone $\sigma_{\parallel} = e^{2}/\pi h$, we have

$$|\varphi_{\max}| = \arctan\pi/2 \approx 1.0039 \text{ rad} \approx 57.5^{\circ}. \tag{49}$$

The above seems difficult to be generalized to the case of a Chern insulator, as the Chern insulator does not have a gapless surface. However, we know there can be bulk Hall current, thus some assumptions have to be modified.

B. Gapped surface with half quantized Hall conductance

We may consider the picture in Fig. 2 (a), where the gapped surface can hold a half-quantized Hall current. In the case, we utilize the local time-reversal symmetry principle, thus the gapless surface can not hold a Hall current. Instead, as shown in the picture, accompanied magnetic or electric fields are generated at the gapless surface, and the gapped

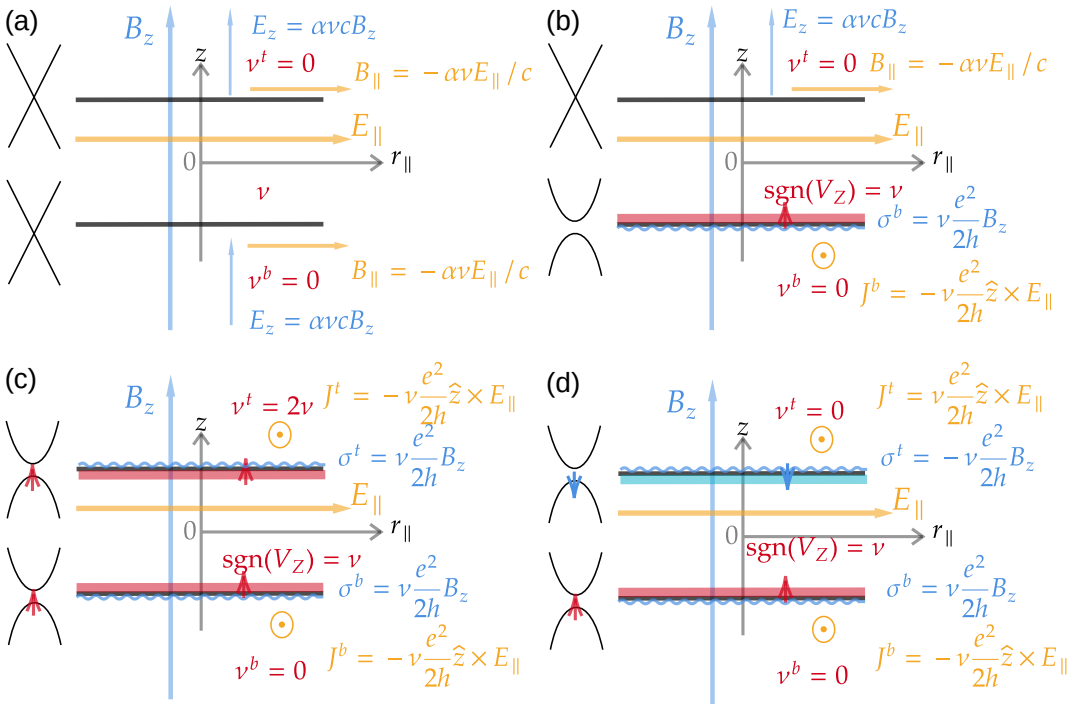


FIG. 2: Schematic diagram of (a) pure TI, (b) half QAHE, (c) QAHE, and (d) axion insulator.

surface holds charge accumulation or a Hall current. We list the relevant boundary conditions as, for a pure TI, $\nu = 0, 1, 0$ in three medium regions,

$$E_z(z > L/2) = \alpha\nu cB_z, \tag{50a}$$

$$E_z(z < L/2) = \alpha\nu cB_z, \tag{50b}$$

$$\boldsymbol{B}_\parallel(z > L/2) = -\alpha\nu\boldsymbol{E}_\parallel/c, \tag{50c}$$

$$\boldsymbol{B}_\parallel(z < L/2) = -\alpha\nu\boldsymbol{E}_\parallel/c; \tag{50d}$$

for the half QAHE, $\nu = 0, 1, 0$ in three medium regions,

$$E_z(z > L/2) = \alpha\nu cB_z, \tag{51a}$$

$$\sigma^b = \nu\frac{e^2}{2h}B_z, \tag{51b}$$

$$\boldsymbol{B}_\parallel(z > L/2) = \alpha\nu\boldsymbol{E}_\parallel/c, \tag{51c}$$

$$\boldsymbol{j}^b = -\nu\frac{e^2}{2h}\hat{z}\times\boldsymbol{E}_\parallel; \tag{51d}$$

for the Chern insulator, $\nu = 2, 1, 0$ in three medium regions,

$$\sigma^t = \nu\frac{e^2}{2h}B_z, \tag{52a}$$

$$\sigma^b = \nu\frac{e^2}{2h}B_z, \tag{52b}$$

$$\boldsymbol{j}^t = -\nu\frac{e^2}{2h}\hat{z}\times\boldsymbol{E}_\parallel, \tag{52c}$$

$$\boldsymbol{j}^b = -\nu\frac{e^2}{2h}\hat{z}\times\boldsymbol{E}_\parallel; \tag{52d}$$

for the axion insulator, $\nu = 0, 1, 0$ in three medium regions,

$$\sigma^t = -\nu\frac{e^2}{2h}B_z, \tag{53a}$$

$$\sigma^b = \nu\frac{e^2}{2h}B_z, \tag{53b}$$

$$\boldsymbol{j}^t = \nu\frac{e^2}{2h}\hat{z}\times\boldsymbol{E}_\parallel, \tag{53c}$$

$$\boldsymbol{j}^b = -\nu\frac{e^2}{2h}\hat{z}\times\boldsymbol{E}_\parallel. \tag{53d}$$

In this consideration, the gapped surface is described by the Chern-Simons theory, with charge and current accumulation, while the gapless surface is not, with generated fields to compensate the boundary condition. For the pure TI case, the Chern-Simons theory is not applicable on both surfaces.

C. Allowing bulk Hall current only

Now we consider that the bulk can hold a Hall current, but can not hold a charge accumulation. We also assume in this section that the surface can not hold a Hall current, but can hold a charge accumulation. In this case, the charge generation shares no difference from the previous case, but the current generation has to be modified.

For illustration, we take $E_x \neq 0$, $E_y = 0 = B_z$ in the whole space, protected by Faraday's law and magnetic Gauss's law. We denote ν to take values ν^t, ν^{TI}, ν^b in three medium regions $z > L/2$, $|z| < L/2$ and $z < -L/2$, with possibilities that $\nu^t = 0, 2$, $\nu^{TI} = 1$ and $\nu^b = 0$, which are connected by the Heaviside step function at the boundary. For our purpose, we need to consider the modified Ampere's law Eq. (33d) in the whole space.

We first consider the z-component of the equation

$$\partial_x B_y - \partial_y B_x - \alpha\left[\partial_x(\nu E_y) - \partial_y(\nu E_x)\right] = \mu_0 J_z, \tag{54}$$

and in the stationary case, we have $J_z \equiv 0$, while by translational invariance, all $\partial_{x/y}\cdot$ to some physical quantity should vanish, thus this component of the equation is automatically

satisfied. With similar consideration, for the other two components, we have

$$-\partial_z B_y + \alpha\partial_z(\nu E_y) = \mu_0 J_x, \tag{55a}$$

$$\partial_z B_x - \alpha\partial_z(\nu E_x) = \mu_0 J_y, \tag{55b}$$

which are differential equations along z-direction.

We first consider regions $|z| > L/2$, where $J_{x/y} \equiv 0$, this directly leads to $\partial_z(\boldsymbol{B}_\parallel - \alpha\nu^{t/b}\boldsymbol{E}_\parallel) = 0$, which gives

$$\boldsymbol{B}_\parallel(z > L/2) = const., \tag{56a}$$

$$\boldsymbol{B}_\parallel(z < L/2) = const.. \tag{56b}$$

Notice that they can be different constants in the two regions.

Next, we consider the boundary conditions at $z = \pm L/2$, and we have, following the same procedure as before,

$$\hat{z} \times \Delta\tilde{\boldsymbol{B}}_\parallel(\pm L/2) = \mu \boldsymbol{j}^{t/b}, \tag{57}$$

where the surface current reads

$$\boldsymbol{j}^{t/b} = \int_{\pm L/2-\varepsilon}^{\pm L/2+\varepsilon} \mathrm{d}z \boldsymbol{J}(z). \tag{58}$$

However, in our case, we allow only a bulk Hall current, and by a non-vanishing E_x, we have $j_y^{t/b} = 0$, which leads to

$$\Delta B_x(L/2) - \alpha(\nu^t - \nu^{TI})E_x = 0, \tag{59a}$$

$$\Delta B_x(-L/2) + \alpha(\nu^b - \nu^{TI})E_x = 0, \tag{59b}$$

which can be written as

$$B_x(L^-/2) = \alpha(\nu^{TI} - \nu^t)E_x + B_x(L^+/2), \tag{60a}$$

$$B_x(-L^+/2) = \alpha(\nu^{TI} - \nu^b)E_x + B_x(-L^-/2), \tag{60b}$$

for later use. One is aware that under the assumption that there is no Hall current on the surface, the parallel magnetic field is unavoidable to be generated, either inside or outside the surface. On the other hand, for $j_x^{t/b}$, it is possible to gain a non-vanishing value through the gapless surface as a longitudinal current, thus generally we write

$$j_x^{t/b} = -\Delta B_y(\pm L/2)/\mu_0, \tag{61}$$

with $E_y = 0$ utilized. This is a result from the usual Ampere's law.

Finally, we take look at the TI bulk, where $J_y \neq 0$ but $J_x = 0$. We thus have

$$-\partial_z B_y = 0, \tag{62a}$$

$$\partial_z B_x = \mu_0 J_y(z). \tag{62b}$$

The solution to Eq. (62b) is constrained by the total Hall conductance of the magnetic TI $\sigma_{xy} = n\frac{e^2}{2h}$ where $n = 0, 1, 2$ for cases, and by noticing that the left-hand-side of Eq. (62b) is a differential form, we can write

$$\begin{aligned} j_y^{bulk} &= \int_{-L/2+\varepsilon}^{L/2-\varepsilon} \mathrm{d}z J_y(z) = \frac{1}{\mu_0}\int_{-L/2+\varepsilon}^{L/2-\varepsilon} \mathrm{d}z \partial_z B_x \\ &= \frac{1}{\mu_0}\left[B_x(L/2-\varepsilon) - B_x(-L/2+\varepsilon)\right] \\ &= -\frac{\alpha}{\mu_0 c}(\nu^t - \nu^b)E_x + \frac{1}{\mu_0}\left[B_x(L/2+\varepsilon) - B_x(-L/2-\varepsilon)\right] \\ &= \sigma_{yx}E_x = -n\frac{e^2}{2h}E_x. \end{aligned} \tag{63}$$

From this equation, we immediately realize that the equation can be solved without knowing the actual current distributions in the bulk, but only the boundary conditions. This means that under the theory of axion electrodynamics, the bulk Hall current distribution is not unique, but the total current can be fixed by the boundary. It is also intriguing that the bulk Hall current does not depend on the ν^{TI} value.

Now we can study case-by-case:

• For a pure TI, we consider $n = 0$, $\nu^t = 0 = \nu^b$, thus $j_y^{bulk} = 0$ leads to $B_x(L/2+\varepsilon) = B_x(-L/2-\varepsilon)$, and further considering Eq. (60) leads to $B_x(|z| > L/2) = B_x(|z| < L/2) - \alpha\nu^{TI}E_x/c$;

• For a half QAHE, we have $n = 1$, and by letting $\nu^t = 0 = \nu^b$, thus $j_y^{bulk} = -\frac{e^2}{2h}E_x$ leads to $B_x(L/2+\varepsilon) = B_x(-L/2-\varepsilon) - \alpha E_x/c$;

• For a Chern insulator, we have $n = 2$, and by letting $\nu^t = 2, \nu^b = 0$, $j_y^{bulk} = -\frac{e^2}{h}E_x$ is directly satisfied;

• For an axion insulator, we also have $n = 0$, $\nu^t = 0 = \nu^b$, thus without surface longitudinal current, it should be indistinguishable from a pure TI.

D. A combination between surface and bulk

Consider most generally the situation that both bulk and surface Hall current is allowed. For the case $E_x \neq 0$ and $E_y = 0$, we have

$$j_y^t = \frac{1}{\mu_0}\Delta B_x(L/2) - \frac{e^2}{2h}(\nu^t - \nu^{TI})E_x, \tag{64a}$$

$$j_y^b = \frac{1}{\mu_0}\Delta B_x(-L/2) + \frac{e^2}{2h}(\nu^b - \nu^{TI})E_x, \tag{64b}$$

$$j_y^{bulk} = \frac{1}{\mu_0}\left[B_x(L/2-\varepsilon) - B_x(-L/2+\varepsilon)\right], \tag{64c}$$

and the total Hall current is

$$\begin{aligned} j_y &= j_y^t + j_y^b + j_y^{bulk} \\ &= -\frac{e^2}{2h}(\nu^t - \nu^b)E_x + \frac{1}{\mu_0}\left[B_x(L/2+\varepsilon) - B_x(-L/2-\varepsilon)\right], \end{aligned} \tag{65}$$

which is the same as the current formula for the bulk Hall current we deduced before (if we preserve all B fields, the result of pure surface current should also be the same). This result is thought-provoking, as it suggests that the bulk Hall current can be solely determined by the boundary conditions, while the Hall current distribution in the TI is never unique.

We can similarly write down charge density for each part as

$$\sigma^t = \varepsilon_0\Delta E_z(L/2) + \frac{e^2}{2h}(\nu^t - \nu^{TI})B_z, \tag{66a}$$

$$\sigma^b = \varepsilon_0\Delta E_z(-L/2) - \frac{e^2}{2h}(\nu^b - \nu^{TI})B_z, \tag{66b}$$

$$\sigma^{bulk} = \varepsilon_0\left[E_z(L/2-\varepsilon) - E_z(-L/2+\varepsilon)\right], \tag{66c}$$

and the total charge density induced by a B_z field reads as follows

$$\begin{aligned} \sigma &= \sigma^t + \sigma^b + \int_{-L/2+\varepsilon}^{L/2-\varepsilon} \mathrm{d}z\rho(z) \\ &= \frac{e^2}{2h}(\nu^t - \nu^b)B_z + \varepsilon_0\left[E_z(L/2+\varepsilon) - E_z(-L/2-\varepsilon)\right], \end{aligned} \tag{67}$$

and again, we see that the charge density is solely determined by the boundary conditions, while the charge distribution in the TI is never unique.

Nevertheless, when applying the formulas Eq. (65) and Eq. (67) to the four cases, we may need to give additional constraints to the system. For instance, when we consider the charge accumulation, the corresponding charge density should be zero for a gapless surface

(conducted away), and similarly a bulk current can remove the bulk charge, then the only possible place allowing a non-vanishing charge density will be the gapped surface for a stationary system, where the z-current also vanishes. This picture gives strong constraints to our formula, and its result for cases. We take half QAHE as an example with its top layer gapless, and the constraints read

$$\sigma^t = 0 \Longrightarrow \varepsilon_0 \Delta E_z(L/2) = -\frac{e^2}{2h}(\nu^t - \nu^{TI})B_z = \frac{e^2}{2h}B_z, \tag{68a}$$

$$\sigma^b = \varepsilon_0 \Delta E_z(-L/2) - \frac{e^2}{2h}(\nu^b - \nu^{TI})B_z = \varepsilon_0 \Delta E_z(-L/2) + \frac{e^2}{2h}B_z, \tag{68b}$$

$$\sigma^{bulk} = 0 \Longrightarrow E_z(L/2+\varepsilon) = E_z(-L/2-\varepsilon), \tag{68c}$$

with $\nu^t = 0 = \nu^b$, $\nu^{TI} = 1$ assumed. At this time, we have some flexibilities introduced by the E_z field. For instance, by choosing downside field $E_z(-L/2-\varepsilon) = 0$, we can have $\sigma^b = \frac{e^2}{2h}B_z + \varepsilon_0 E_z^{TI}$, where we use E_z^{TI} to denote the homogeneous electric field in TI bulk, and the top surface constraint can be satisfied by $\varepsilon_0 E_z(L/2+\varepsilon) = \varepsilon_0 E_z^{TI} + \frac{e^2}{2h}B_z$, which can take arbitrary values. However, by assuming that electric field in TI is zero, i.e., $E_z^{TI} = 0$, we can have $\sigma^b = \frac{e^2}{2h}B_z$ and $E_z(L/2+\varepsilon) = \alpha c B_z$, which is consistent with the previous results. Other cases follow from similar consideration, and the results are consistent with the previous results.

E. A semi-infinite structure

Now let us try to consider the case where TI lies at lower half space with boundary at $z = 0$, and one can write equations for top surface and bulk as

$$j_y^t = \frac{1}{\mu_0}\Delta B_x(0) - \frac{e^2}{2h}(\nu^t - \nu^{TI})E_x, \tag{69a}$$

$$j_y^{bulk} = \frac{1}{\mu_0}\left[B_x(0^-) - B_x(-\infty)\right], \tag{69b}$$

$$j_y = \frac{1}{\mu_0}\left[B_x(0^+) - B_x(-\infty)\right] - \frac{e^2}{2h}(\nu^t - \nu^{TI})E_x; \tag{69c}$$

$$\sigma^t = \varepsilon_0 \Delta E_z(0) + \frac{e^2}{2h}(\nu^t - \nu^{TI})B_z, \tag{70a}$$

$$\sigma^{bulk} = \varepsilon_0\left[E_z(0^-) - E_z(-\infty)\right], \tag{70b}$$

$$\sigma = \varepsilon_0 \left[E_z(0^+) - E_z(-\infty)\right] + \frac{e^2}{2h}(\nu^t - \nu^{TI})B_z. \tag{70c}$$

We know that when the surface is gapless, the system will contribute totally a half-quantized Hall conductance (we may be careful and willing to consider one Dirac cone), and the half-quantization should be a bulk contribution, which motivates us to write

$$j_y^t = \frac{1}{\mu_0}\Delta B_x(0) - \frac{e^2}{2h}(\nu^t - \nu^{TI})E_x = 0, \tag{71a}$$

$$j_y^{bulk} = \frac{1}{\mu_0}\left[B_x(0^-) - B_x(-\infty)\right] = S\frac{e^2}{2h}E_x, \tag{71b}$$

$$\sigma^t = \varepsilon_0 \Delta E_z(0) + \frac{e^2}{2h}(\nu^t - \nu^{TI})B_z = 0, \tag{71c}$$

$$\sigma^{bulk} = \varepsilon_0 \left[E_z(0^-) - E_z(-\infty)\right] = 0, \tag{71d}$$

which leads to, by assuming $\nu^t = 0, \nu^{TI} = 1$, $B_x(\pm\infty) = 0 = E_z(-\infty)$,

$$\begin{cases} B_x(0^-) = \alpha E_x/c \\ S = +1 \\ E_z(0^+) = \alpha c B_z \end{cases}. \tag{72}$$

On the other hand, for TI with a gapped surface, Hall is zero while surface charge is non-zero, and we have

$$j_y^t = \frac{1}{\mu_0}\Delta B_x(0) - \frac{e^2}{2h}(\nu^t - \nu^{TI})E_x = 0, \tag{73a}$$

$$j_y^{bulk} = \frac{1}{\mu_0}\left[B_x(0^-) - B_x(-\infty)\right] = 0, \tag{73b}$$

$$\sigma^t = \varepsilon_0 \Delta E_z(0) + \frac{e^2}{2h}(\nu^t - \nu^{TI})B_z, \tag{73c}$$

$$\sigma^{bulk} = \varepsilon_0 \left[E_z(0^-) - E_z(-\infty)\right] = 0, \tag{73d}$$

which leads to, by assuming $\nu^t = 0, \nu^{TI} = 1$, $B_x(-\infty) = 0 = E_z(\pm\infty)$,

$$\begin{cases} B_x(0^+) = -\alpha E_x/c \\ \sigma^t = \frac{e^2}{2h}(\nu^t - \nu^{TI})B_z \end{cases}. \tag{74}$$

IV. Side Surface Half Quantized Anomalous Hall Effect by Current

A. Homogeneous in-layer gate

$$\sigma_{xy} = \frac{j_x}{\mu/e}, \tag{75a}$$

$$\sigma_{xz} = \frac{j_x}{(G_t - G_b)/e}. \tag{75b}$$

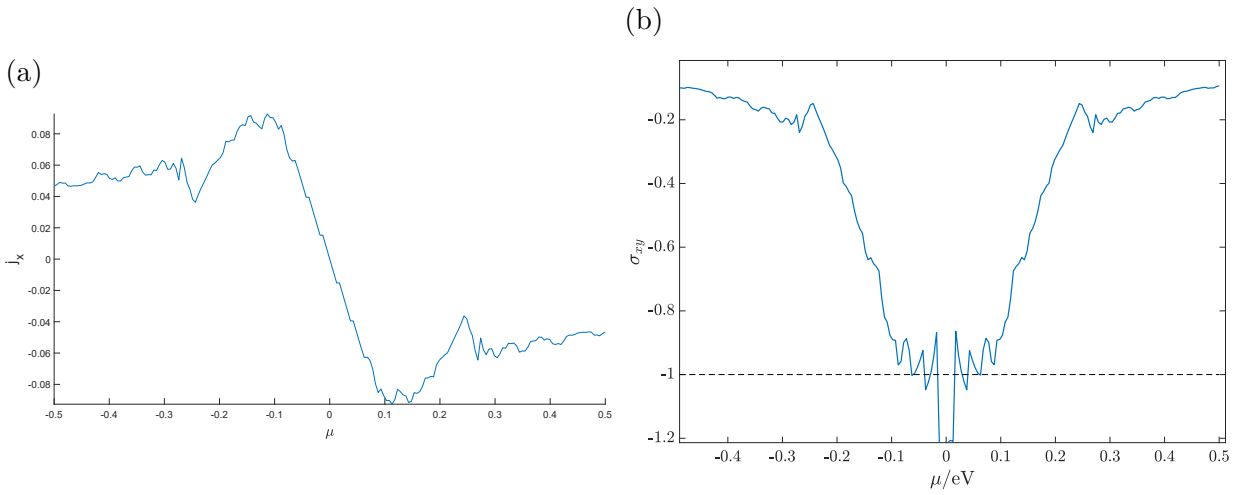


FIG. 3: (a) j_x. (b) σ_{xz} for left side-surface. $L_z = 20$, $L_y = 30$, there are 400 k-points, layer-homogeneous gate $G_{t/b} = \pm 0.01$ eV, $V_{t/b} = 0.1$ eV for 2-layer thickness as an Chern insulator configuration.

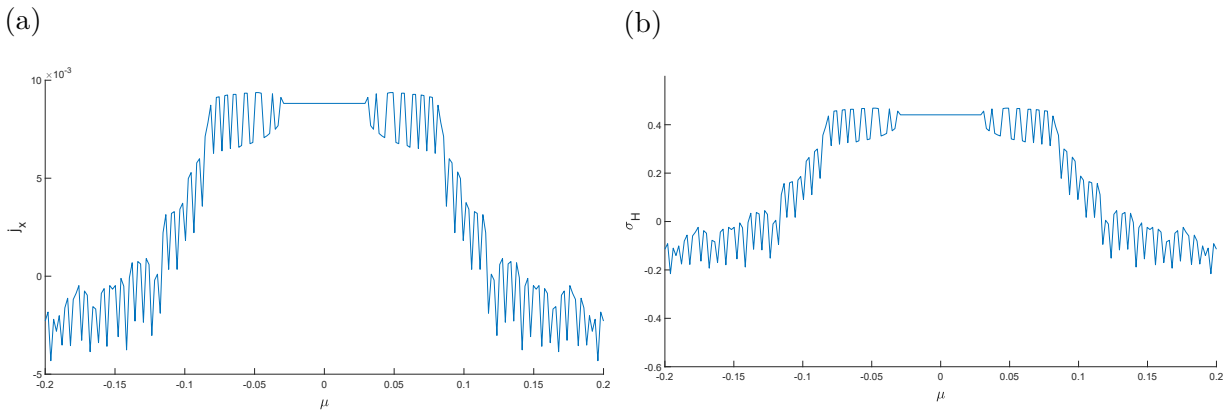


FIG. 4: (a) j_x. (b) σ_{xz} for left side-surface. $L_z = 20$, $L_y = 30$, there are 800 k-points,, layer-homogeneous gate $G_{t/b} = \pm 0.01$ eV, $V_{t/b} = \pm 0.1$ eV for 2-layer thickness as an axion insulator configuration.

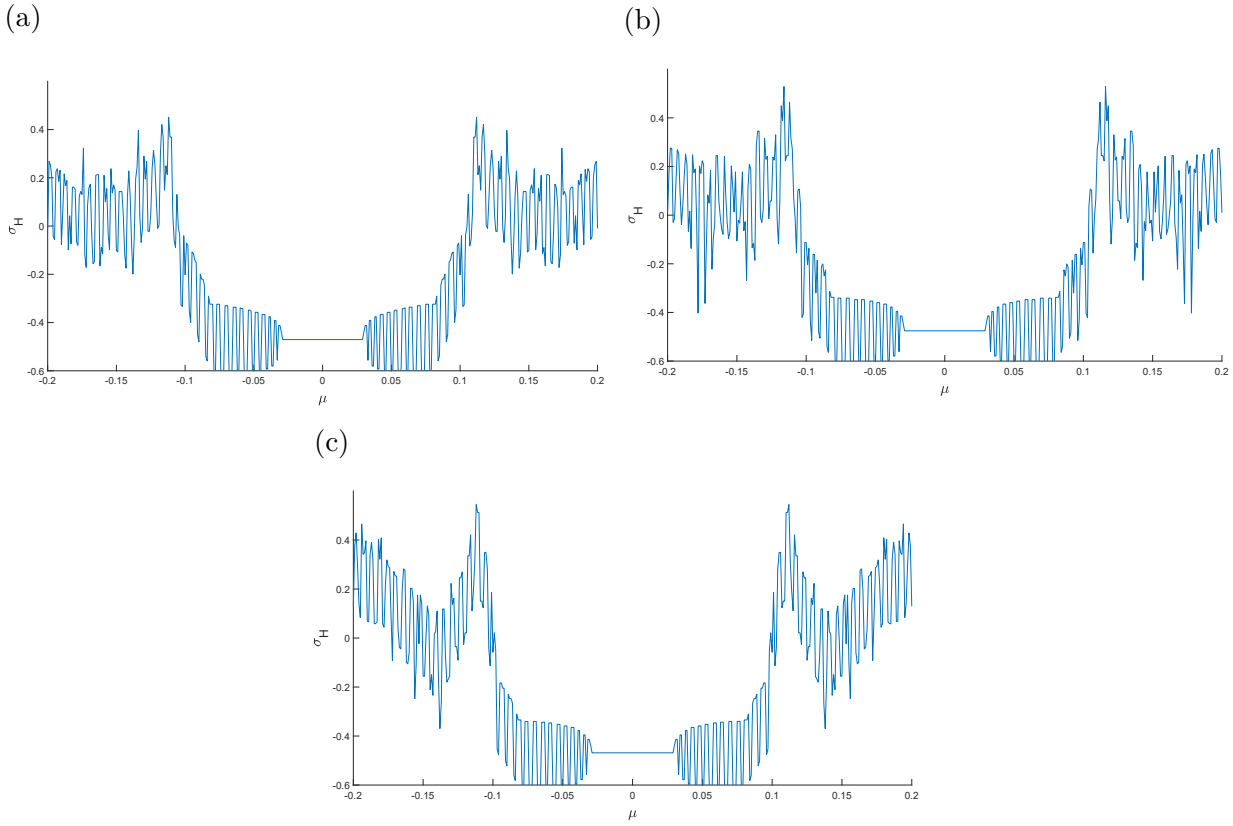


FIG. 5: σ_{xz} for right side-surface. $L_z = 20$, $L_y = 30$, there are 400 k-points, layer-homogeneous gate $G_{t/b} = \pm 0.01$ eV, $V_{t/b} = \pm 0.1$ eV for 2-layer thickness. Additionally, a y-Zeeman field is added at the left side-surface with directions as (a) Left-out. (b) Left-in. (c) Left-out with large Zeeman.

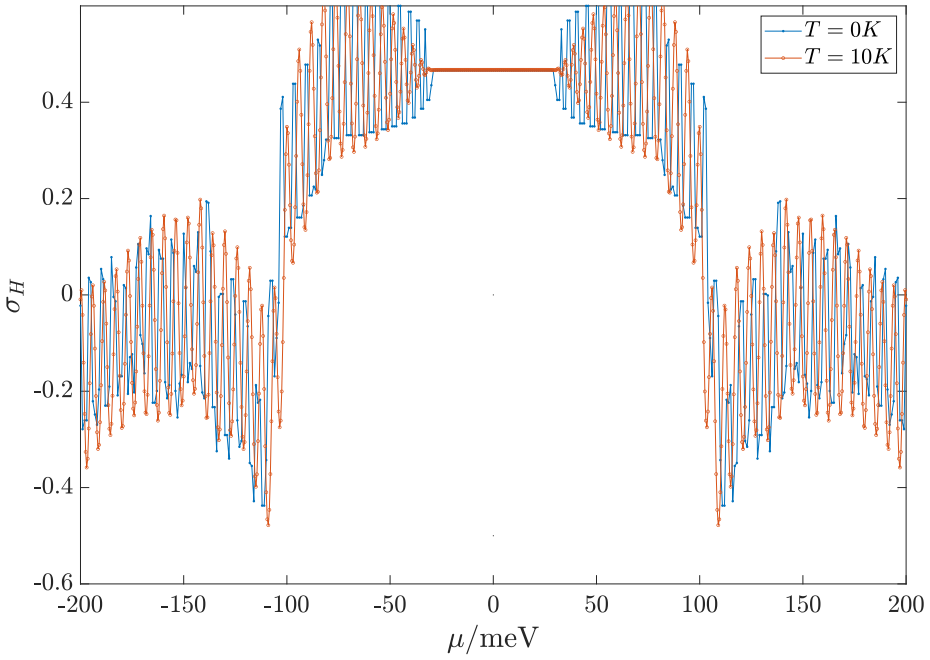


FIG. 6: σ_{xz} for left side-surface at $T = 0$ and $T = 10$ K. $L_z = 20$, $L_y = 20$, there are 400 k-points, layer-homogeneous gate $G_{t/b} = \pm 1 \times 10^{-3}$ eV, $V_{t/b} = \pm 0.1$ eV for 2-layer thickness as an axion insulator configuration.

B. Some layer Hall conductance

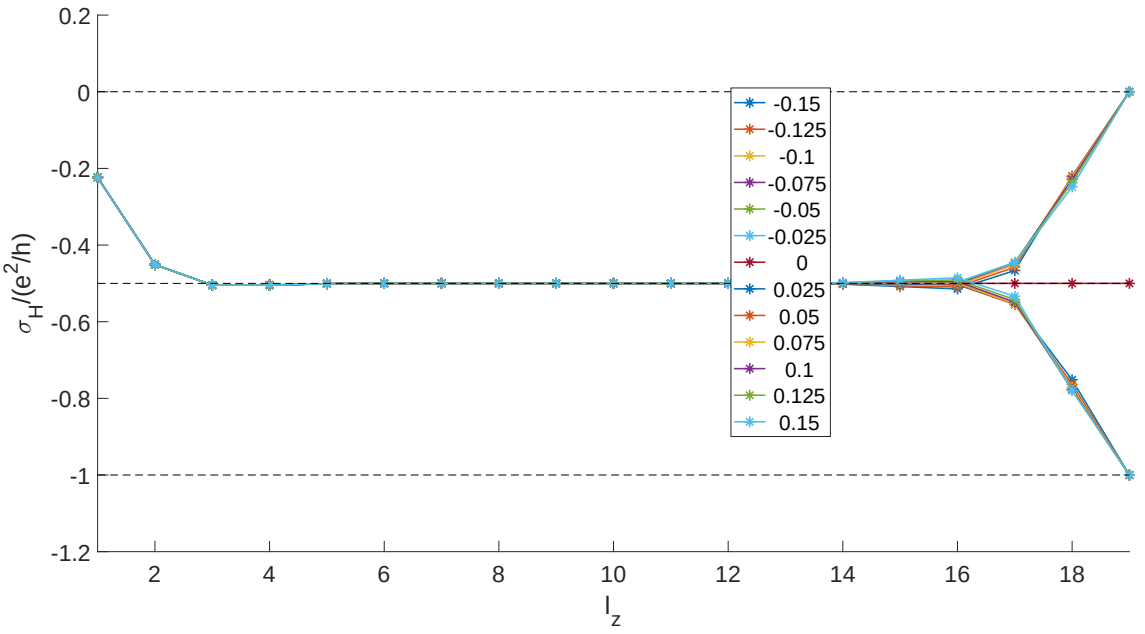


FIG. 7: Accumulated layer Hall conductance $\tilde{\sigma}_{xy}(l_z) = \sum_{j<l_z} \sigma_{xy}(j)$ with fixed top Zeeman $V_t = 0.1$ eV and varying bottom Zeeman field immersed in 2 layers. $L_z = 19$.

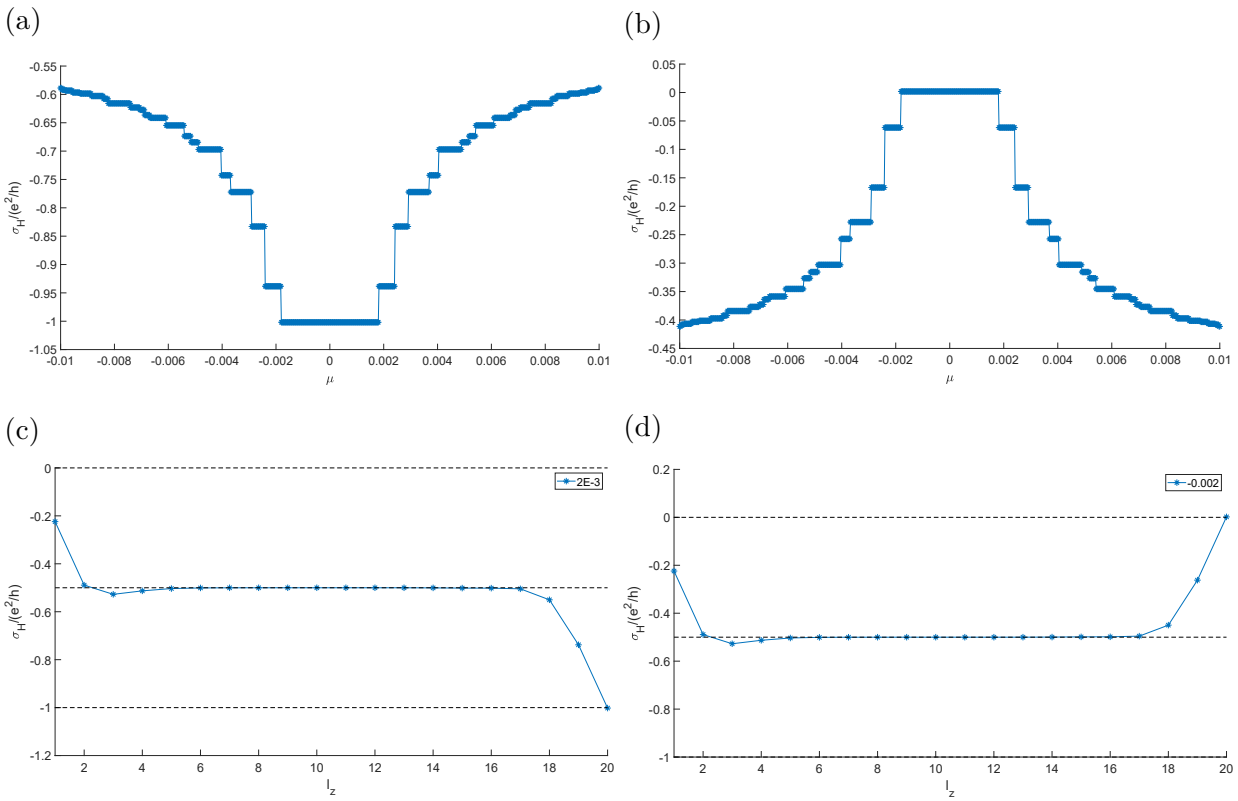


FIG. 8: Hall conductance varying with chemical potential (a) (b), and accumulated layer Hall conductance varying with layer index (c) (d). (a) (c) for Chern insulator configuration with $V_t = 0.2$ eV and $V_b = 2 \times 10^{-3}$ eV, and (b) (d) for axion insulator configuration with $V_t = 0.2$ eV and $V_b = 2 \times 10^{-3}$ eV. $L_z = 20$, k-points are 1600×1600.

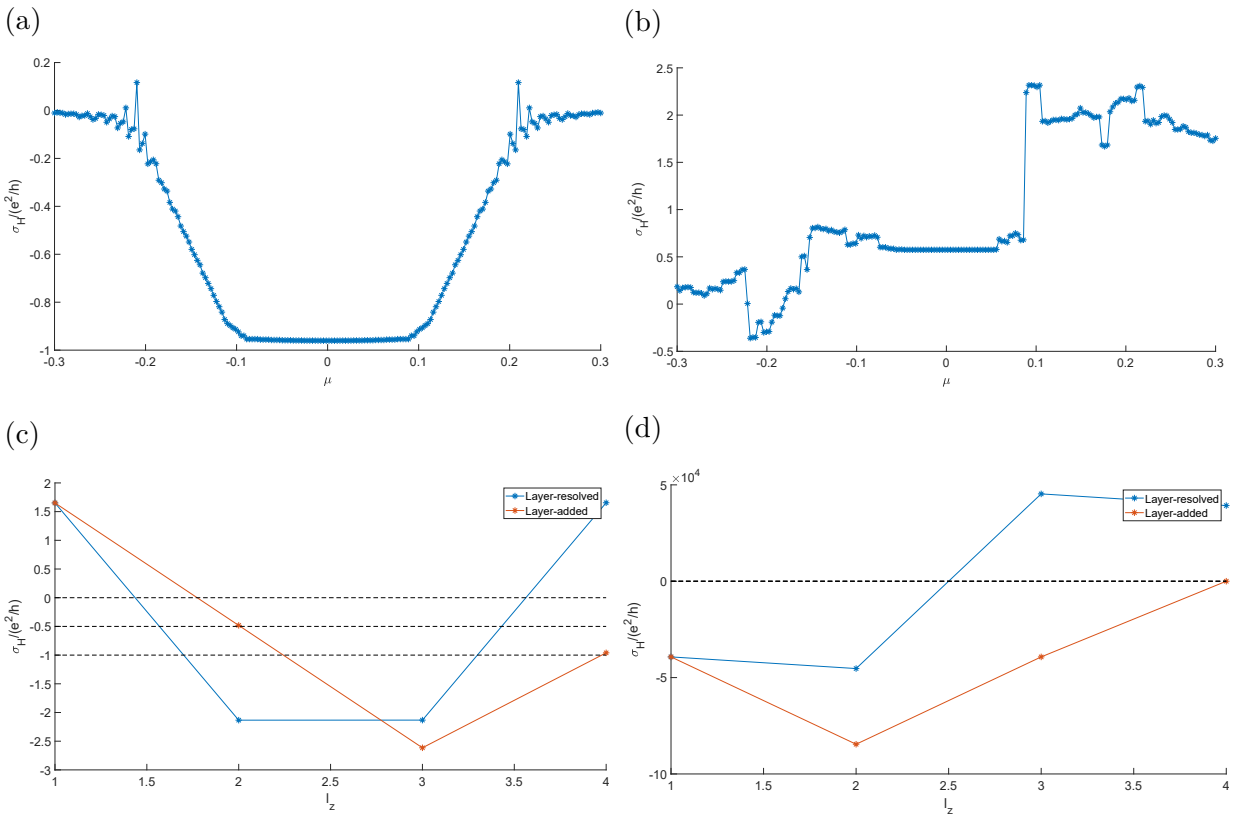


FIG. 9: Hall conductance varying with chemical potential (a) (b), and accumulated layer Hall conductance varying with layer index (c) (d). (a) (c) for Chern insulator configuration with homogeneous $V_z = 0.1$ eV, and (b) (d) for axion insulator configuration with $V_t = 0.1$ eV and $V_b = -0.1$ eV immersed in 1 layer. $L_z = 4$, $L_y = 80$, k_x-points are 200.

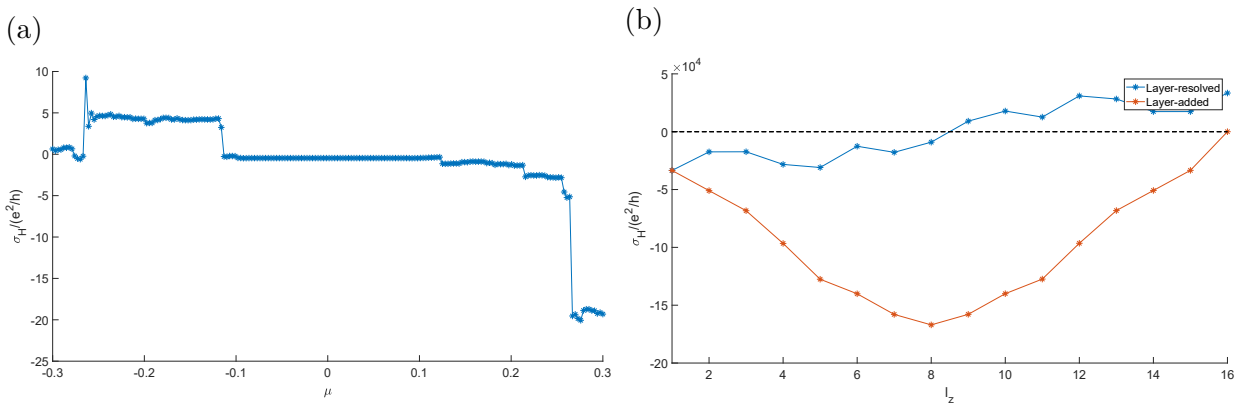


FIG. 10: Hall conductance varying with chemical potential (a) (b), and accumulated layer Hall conductance varying with layer index (c) (d). (a) (c) for Chern insulator configuration with homogeneous $V_z = 0.1$ eV, and (b) (d) for axion insulator configuration with $V_t = 0.1$ eV and $V_b = -0.1$ eV immersed in 2 layer. $L_z = 16$, $L_y = 80$, k_x-points are 200.

C. In-layer gate at the corner

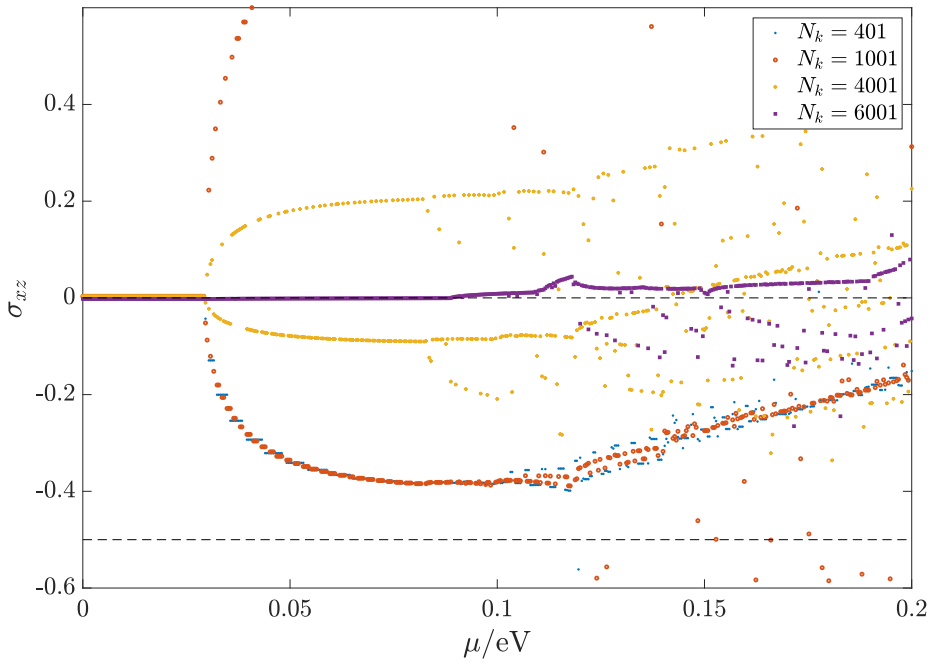


FIG. 11: σ_{xz} for left side-surface varying with k-mesh density at $T = 0K$. $L_z = 20$, $L_y = 20$, in-layer gate $G_{t/b} = \pm 0.01$ eV are added at corner 4-layers, $V_{t/b} = \pm 0.1$ eV for 2-layer thickness as an axion insulator configuration, and a y-Zeeman field is added at the right side-surface with direction as in.

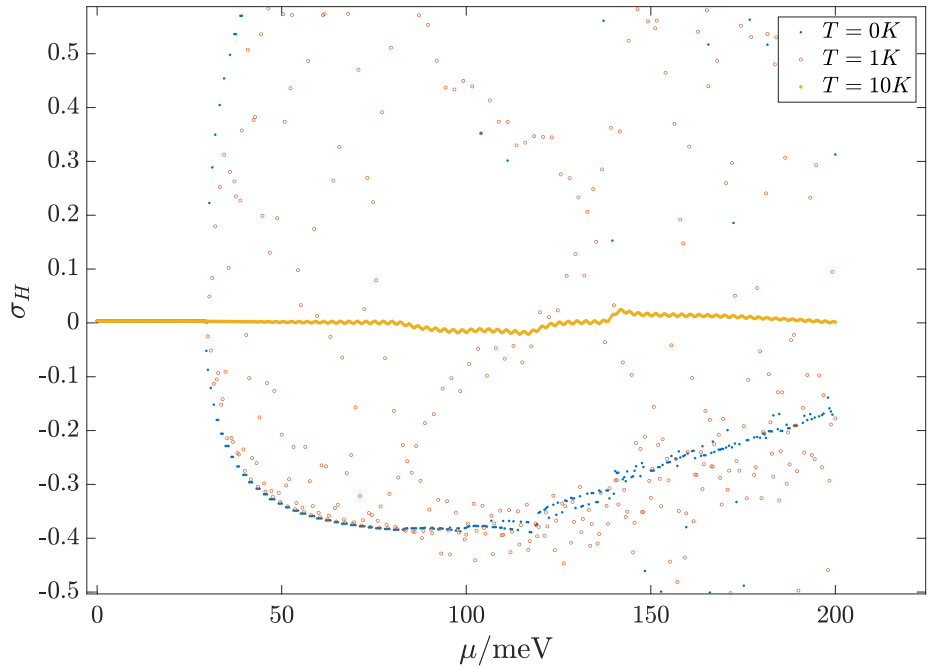


FIG. 12: σ_{xz} for left side-surface varying with temperature. $L_z = 20$, $L_y = 20$, k-points are 1000×1000, in-layer gate $G_{t/b} = \pm 0.01$ eV are added at corner 4-layers, $V_{t/b} = \pm 0.1$ eV for 2-layer thickness as an axion insulator configuration, and a y-Zeeman field is added at the right side-surface with direction as in.

D. When four side surfaces are all gapped

1. When are there hinge states?

A first argument in our model that the same out-Zeeman fields on neighboring surfaces will lead to a chiral hinge mode comes as follows. Without loss of generality, we consider the top surface at $z = 0$ and the right surface at $y = 0$, and the surface state Hamiltonian at low energies reads (assuming isotropy)

$$H_{\text{top}} = v_F(k_x\sigma_y - k_y\sigma_x) + V_t\sigma_z, \tag{76a}$$

$$H_{\text{right}} = v_F(k_z\sigma_x - k_x\sigma_z) + V_r\sigma_y, \tag{76b}$$

and to see the hinge mode, we rotate the right surface Hamiltonian by $\pi/2$ along x-axis by $U_x^{\pi/2} = \exp(iS_x\pi/2\hbar)$ with $S_x = \hbar\sigma_x/2$ as

$$\tilde{H}_{\text{right}}(k_x, k_y) = U_x^{\pi/2\dagger} H_{\text{right}}(M_x^{\pi/2}(k_z, k_x))U_x^{\pi/2\dagger} = v_F(-k_y\sigma_x + k_x\sigma_y) + V_r\sigma_z, \tag{77}$$

with replacement $M_x^{\pi/2}(k_z, k_x) = (-k_y, k_x)$. Immediately, we see that the hinge mode is formed, as a domain-wall mode, by the top and right surface states when V_t and V_r have the opposite signs. Inversely, when V_t and V_r have the same sign, the hinge mode is absent. Further consider the bottom Hamiltonian

$$H_{\text{bottom}} = -v_F(k_x\sigma_y - k_y\sigma_x) + V_b\sigma_z, \tag{78}$$

and to make it consistent with the top surface, we rotate it by π along x-axis using $U_x^{\pi} = \exp(iS_x\pi\hbar)$, and

$$\tilde{H}_{\text{bottom}}(k_x, k_y) = U_x^{\pi\dagger} H_{\text{bottom}}(M_x^{\pi}(k_x, k_y))U_x^{\pi\dagger} = v_F(k_x\sigma_y - k_y\sigma_x) - V_b\sigma_z, \tag{79}$$

where the net effect is an additional minus sign in the V_b term, and a chiral boundary mode is formed when V_t and V_b have the same signs, as in the case of a Chern insulator.

2. Chern insulator

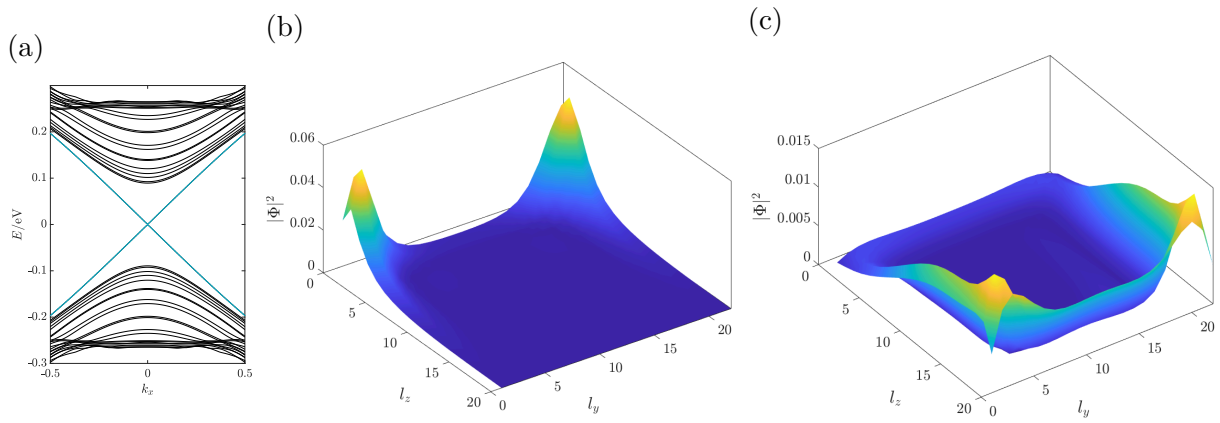


FIG. 13: (a) Band structure for a Chern insulator with two side surfaces gapped out by Zeeman field. The in-gap states are non-degenerate when $k_x \neq 0$. (b) (c) The lowest and second-lowest wavefunction distribution. $V_{t/b} = 0.1$ eV, $V_{l/r} = \mp 0.1$ eV for 2-layers each, and $L_{z/y} = 20$.

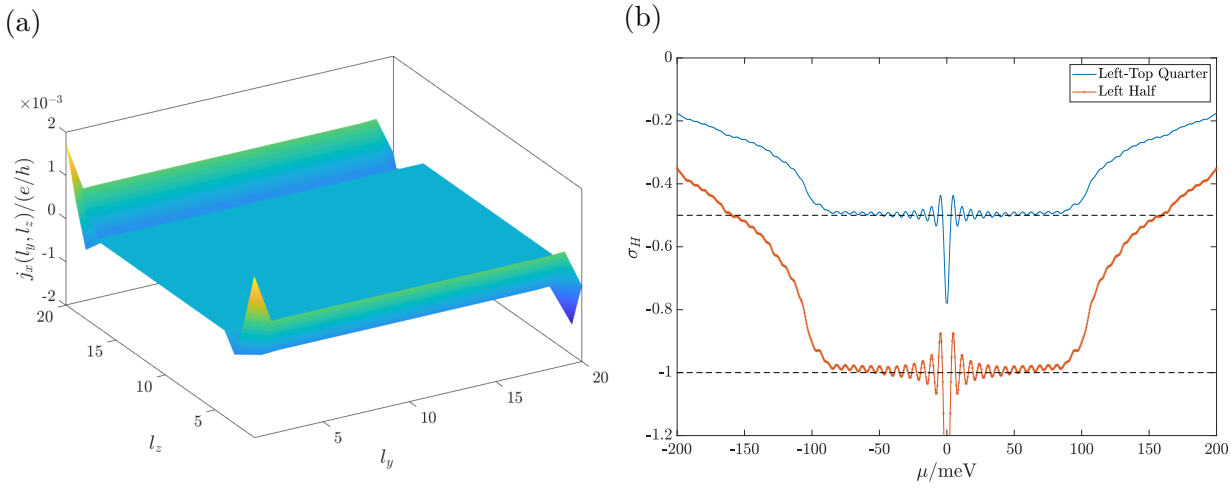


FIG. 14: Corresponding current density (a) and Hall conductance (b) varying with chemical potential. In (a), we set $\mu = 50$ meV, which corresponds a relatively smooth plateau in (b). In (b), we present the Hall conductance for the left-half and the left-top quarter, respectively. We set temperature $T = 10\ K$, which should not affect the result significantly. k-mesh is 400.

3. *Half QAHE*

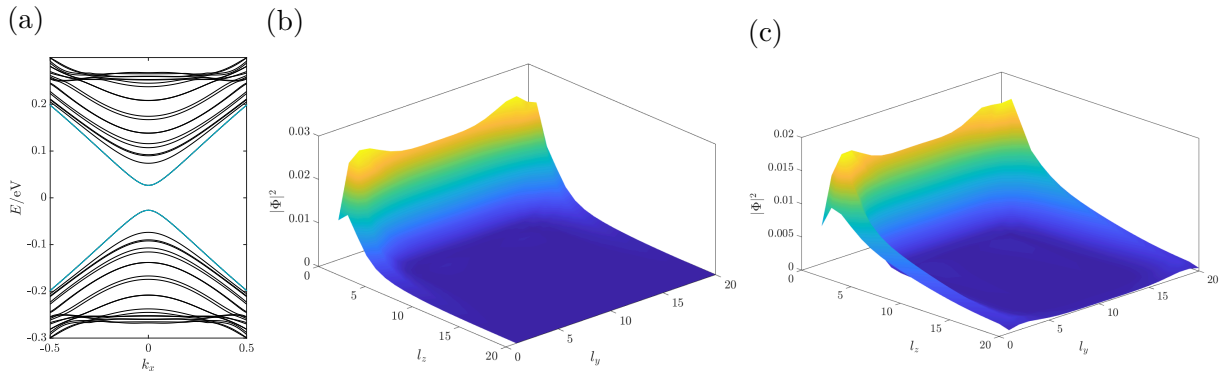


FIG. 15: (a) Band structure for a half-QAHE with two side surfaces gapped out by Zeeman field. The in-gap states are non-degenerate. (b) (c) The lowest and second-lowest wavefunction distribution. $V_t = 0$, $V_b = 0.1$ eV, $V_{l/r} = \mp 0.1$ eV for 2-layers each, and $L_{z/y} = 20$.

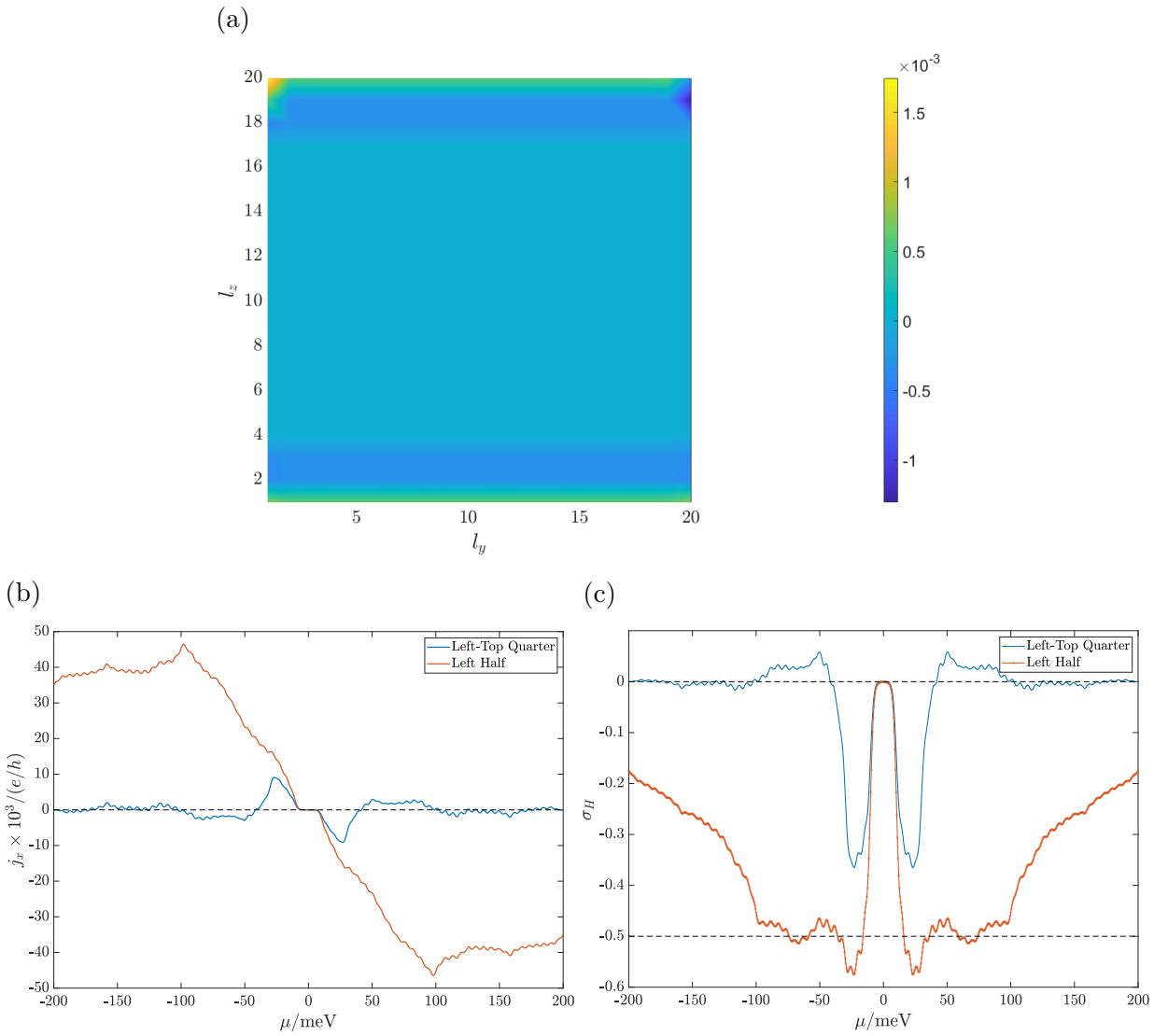


FIG. 16: Corresponding current density (a), current (b) and Hall conductance (c) varying with chemical potential. In (a), we set $\mu = 50$ meV, which corresponds a relatively smooth plateau in (c). In (c), we present the Hall conductance for the left-half and the left-top quarter, respectively. We set temperature $T = 10\ K$, which should not affect the result significantly. k-mesh is 400.

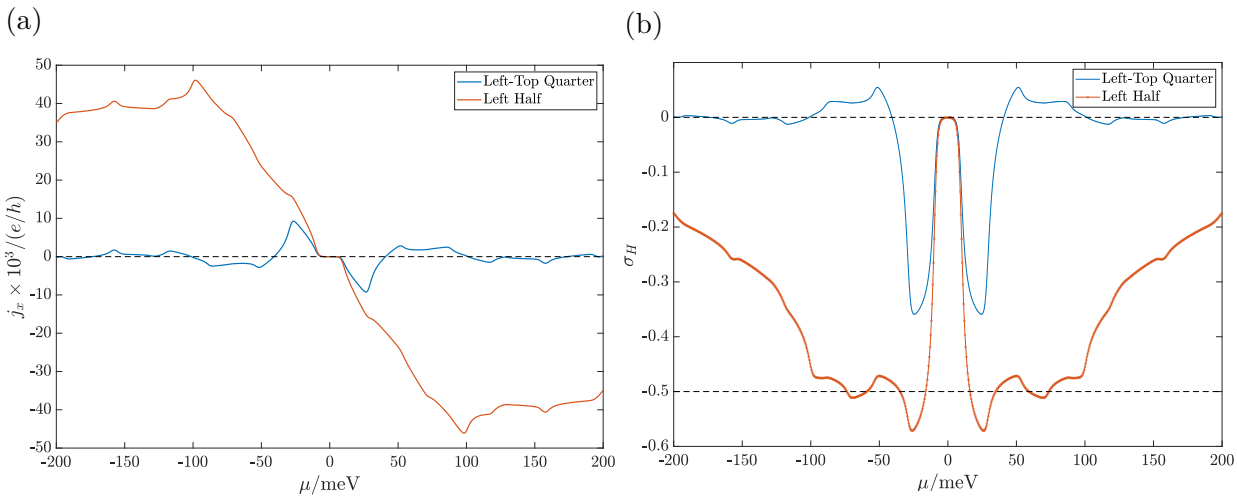


FIG. 17: Corresponding current density (a), current (b) and Hall conductance (c) varying with chemical potential. In (a), we set $\mu = 50$ meV, which corresponds a relatively smooth plateau in (c). In (c), we present the Hall conductance for the left-half and the left-top quarter, respectively. We set temperature $T = 10\ K$, which should not affect the result significantly. k-mesh is 2000.

4. Axion insulator

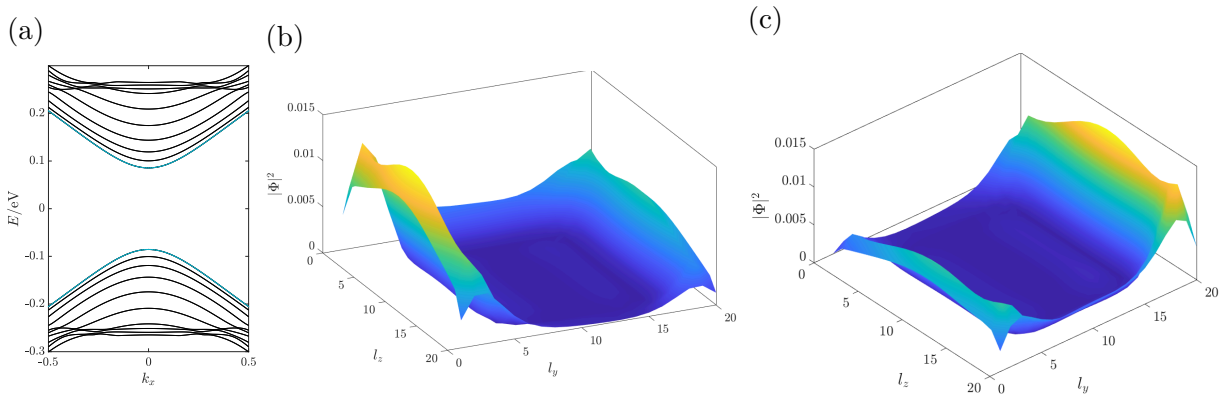


FIG. 18: (a) Band structure for an axion insulator with two side surfaces gapped out by Zeeman field. The in-gap states are doubly degenerate. (b) (c) The lowest and second-lowest wavefunction distribution. $V_{t/b} = \mp 0.1$ eV, $V_{l/r} = \mp 0.1$ eV for 2-layers each, and $L_{z/y} = 20$.

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[4]

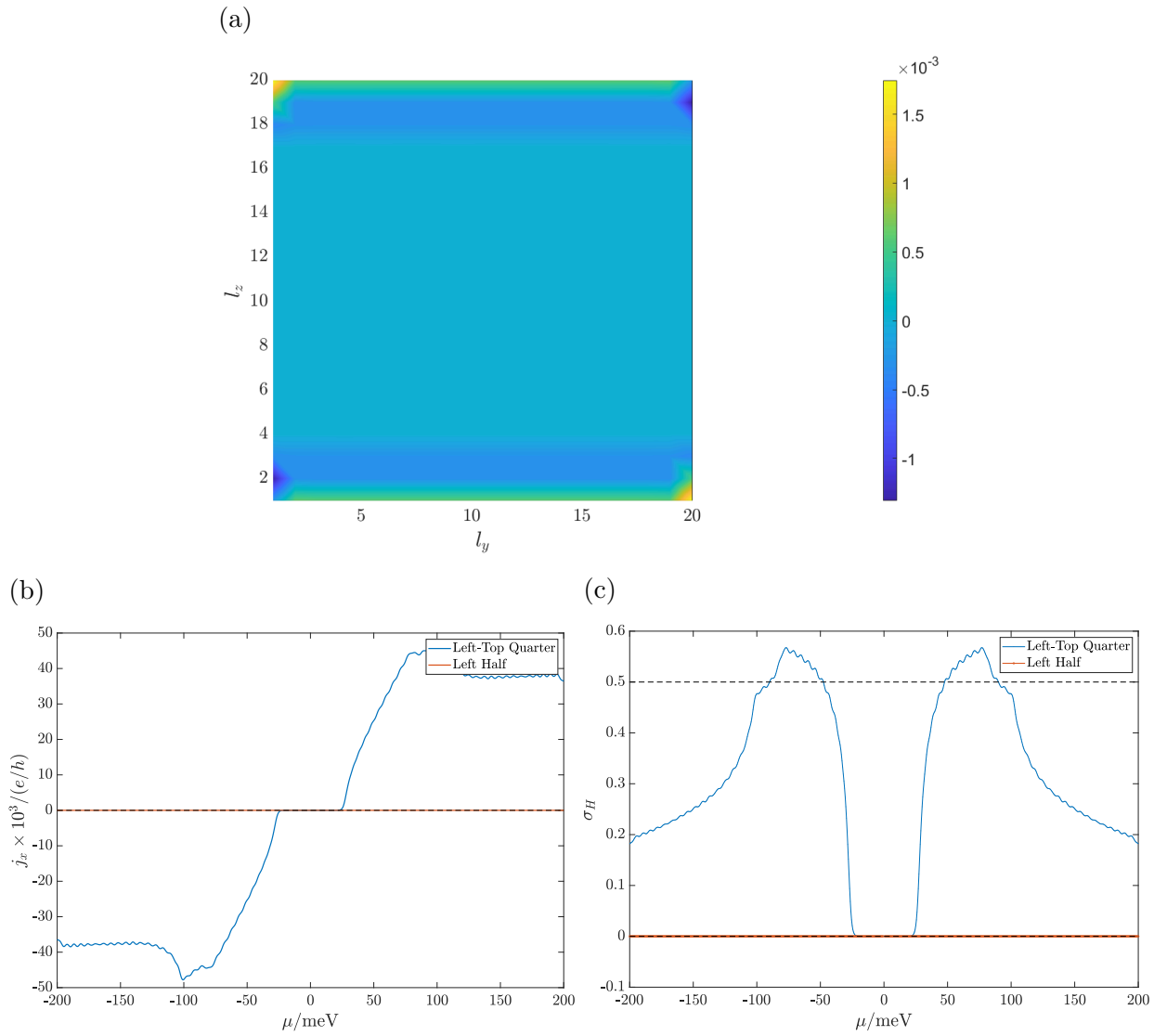


FIG. 19: Corresponding current density (a), current (b) and Hall conductance (c) varying with chemical potential. In (a), we set $\mu = 50$ meV, which corresponds a relatively smooth plateau in (c). In (c), we present the Hall conductance for the left-half and the left-top quarter, respectively. We set temperature $T = 10\ K$, which should not affect the result significantly. k-mesh is 400.